Contour Plots

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Introduction

A contour plot draws isolines — curves along which a scalar field is constant — over a 2D domain. The same geom serves two related purposes: visualising the joint density of paired observations, where the contours are level sets of a kernel-density estimate, and visualising a deterministic function $z = f(x, y)$, where the contours are mathematical isolines. Both uses share a common visual grammar — peaks, ridges, saddles — and contour plots remain one of the most compact ways to communicate where probability mass concentrates or where a function attains specific values.

Prerequisites

A working knowledge of `ggplot2`, basic kernel-density estimation in two dimensions, and the convention that closer contour lines indicate steeper gradients.

Theory

Two distinct workflows feed contour layers in `ggplot2`:

- **Density contours** are computed from raw paired observations via `geom_density_2d()` or `stat_density_2d()`. Internally, a 2D kernel density estimate produces a smoothed surface and `MASS::kde2d()` or `ggplot2`'s built-in estimator returns the level sets.
- **Functional contours** require a complete grid of (x, y, z) values supplied by the user; `geom_contour()` then computes isolines of `z` exactly. The grid resolution determines smoothness; coarse grids produce jagged contours.

Filled variants — `geom_density_2d_filled()` and `geom_contour_filled()` — render closed bands rather than lines and demand a sequential or diverging colour palette so that ordering is preserved.

Assumptions

For density contours, the input is a sample large enough that a kernel density is meaningful (at least a few dozen points; ideally hundreds). For functional contours, the grid covers the domain of interest at adequate resolution.

R Implementation

```r
library(ggplot2); library(metR)

# Contour of 2D density
ggplot(faithful, aes(eruptions, waiting)) +
  geom_point(alpha = 0.3) +
  geom_density_2d(colour = "#F4A261") +
  theme_minimal()

# Filled contour
ggplot(faithful, aes(eruptions, waiting)) +
```

```r
  geom_density_2d_filled(alpha = 0.7) +
  geom_point(alpha = 0.3, size = 0.5, colour = "white") +
  theme_minimal()

# Function surface: z = sin(x) * cos(y)
grid <- expand.grid(x = seq(-3, 3, 0.1), y = seq(-3, 3, 0.1))
grid$z <- sin(grid$x) * cos(grid$y)
ggplot(grid, aes(x, y, z = z)) +
  geom_contour_filled() +
  theme_minimal()
```

Output & Results

Three figures: density contours of the Old Faithful eruption / waiting joint distribution showing the two well-known modes, the same density rendered as filled bands with the underlying points overlaid, and a filled-contour view of $z = \sin(x)\cos(y)$ on a 61×61 grid showing alternating positive and negative regions.

Interpretation

A reporting sentence (figure caption): "Top: 2D kernel density of eruption duration vs waiting time for Old Faithful (n = 272), with raw observations as background points; the bimodality reflects two distinct eruption regimes. Bottom: deterministic surface $z = \sin(x)\cos(y)$ with filled contours at default levels." Mention the kernel and bandwidth choice when density contours are used, since they shape the visible structure.

Practical Tips

- For sparse data, density contours can be misleading — the estimator extrapolates into low-mass regions and produces artefactual closed loops. Use `bins = 5` or `bins = 8` for sparse samples; let the default handle dense ones.
- Always overlay raw points (with low alpha) on density contour plots; the points anchor the contours to actual observations and let readers spot extrapolation.
- For functional surfaces, increase grid resolution until the contours are visibly smooth; under-resolved grids produce stair-stepped lines.
- `metR::geom_contour_fill()` and `metR::geom_text_contour()` add labelled contours that read more clearly on filled backgrounds.
- Use sequential palettes (`viridis`, `magma`) for ordered density and diverging palettes (`RdBu`, `BrBG`) for surfaces that cross zero; qualitative palettes destroy the level ordering and are wrong for filled contours.
- For published figures with multiple contour layers, label a few key isolines with their level values so readers can read off heights without consulting a separate legend.

R Packages Used

`ggplot2` for `geom_density_2d()`, `geom_density_2d_filled()`, `geom_contour()`, and `geom_contour_filled()`; `metR` for `geom_contour_fill()` and labelled contours; `MASS::kde2d()` as the underlying kernel density estimator when finer control is needed.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts